



INCIDENT REPORT

Laboratoire Kastler Brossel

REPORT INFORMATION

Title Loss of Laser Lock During Long Acquisition

Date 15-07-2026

PROBLEM DESCRIPTION

The laser lock was repeatedly lost during acquisitions longer than two hours.

The photodiode signal showed increasing oscillations before the lock failure.

Frequency stability degraded progressively until the control loop became unstable.

ROOT CAUSE

The PID controller gain was configured too aggressively for long-term operation.

Temperature drift modified the response of the piezo actuator and amplified the control loop oscillations.

The issue became visible only during long acquisitions.

CORRECTIVE ACTION

The PID gains were reduced and a temperature compensation procedure was introduced.

The piezo actuator was recalibrated and the lock stability was verified during a six-hour acquisition test.

No further lock losses were observed.